

■ Titanic Exploratory Data Analysis

Comprehensive EDA Report — Kaggle Dataset

Author: Yasodha Krishna Sajja | Date: June 2026 | Tool: Python / Jupyter

1. Project Overview

The Titanic disaster remains one of the most well-known maritime tragedies in history. This project performs an in-depth Exploratory Data Analysis (EDA) on the Titanic dataset to uncover patterns, trends, and relationships that influenced passenger survival. Using Python and data visualisation techniques, demographic characteristics, socioeconomic factors, and travel information were investigated to derive meaningful insights.

Attribute	Detail
Source	Kaggle – Titanic: Machine Learning from Disaster
Dataset	train.csv
Records	891 passengers
Features	12 columns
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter

Objectives

- Understand the structure and quality of the dataset.
- Perform statistical exploration of the data.
- Identify missing values and potential data quality issues.
- Analyse the distribution of important features.
- Investigate relationships between variables and survival outcomes.
- Detect trends, patterns, and anomalies using visualisations.
- Summarise key findings through data-driven insights.

2. Data Understanding

The dataset contains information on **891 passengers** described by **12 features**, encompassing demographic details, ticket information, and survival outcomes. Features include both numerical and categorical variables.

Feature Overview

Feature	Type	Description
PassengerId	Integer	Unique identifier for each passenger
Survived	Integer (0/1)	Survival status: 0 = No, 1 = Yes
Pclass	Integer (1–3)	Ticket/passenger class (socioeconomic proxy)
Name	String	Full name of the passenger
Sex	String	Gender of the passenger
Age	Float	Age in years (177 missing values)
SibSp	Integer	Number of siblings/spouses aboard
Parch	Integer	Number of parents/children aboard
Ticket	String	Ticket number
Fare	Float	Ticket fare paid
Cabin	String	Cabin number (687 missing values)
Embarked	String	Port of embarkation: C, Q, or S

Descriptive Statistics

Feature	Count	Mean	Std Dev	Min	25%	Median	75%	Max
Age	714	29.70	14.53	0.42	20.13	28.00	38.00	80.00
Fare	891	32.20	49.69	0.00	7.91	14.45	31.00	512.33
SibSp	891	0.52	1.10	0	0	0	1	8
Parch	891	0.38	0.81	0	0	0	0	6

Key Insight: The average passenger age was approximately 30 years. Fares exhibit strong right-skew, indicating significant economic diversity. Survival rate was ~38.4%, meaning the majority of passengers did not survive.

3. Data Quality Checks & Cleaning

Real-world datasets contain inconsistencies such as missing values, duplicate records, and incorrect data types. Since the objective is EDA rather than predictive modelling, only minimal cleaning was applied to preserve data authenticity.

Missing Value Assessment

Feature	Missing Values	Percentage (%)	Treatment Strategy
Cabin	687	77.10%	Retained as-is — too sparse to impute reliably
Age	177	19.87%	Median imputation to reduce outlier influence
Embarked	2	0.22%	Mode imputation (negligible missingness)

- No duplicate records were found — each row represents a unique passenger.
- All data types were verified as appropriate for analysis.
- Numerical variables: Age, Fare, SibSp, Parch. Categorical: Sex, Embarked, Pclass.
- Outliers in Age and Fare were retained as they represent genuine passenger characteristics.

4. Univariate Analysis

Univariate analysis examines individual variables independently to understand their distributions, central tendencies, and potential anomalies.

4.1 Survival Distribution

Status	Count	Percentage
Did Not Survive (0)	549	61.62%
Survived (1)	342	38.38%

Key Insight: Approximately 62% of passengers did not survive. The dataset exhibits class imbalance toward non-survivors, which must be accounted for in any future predictive modelling.

4.2 Gender Distribution

Gender	Count	Percentage
Male	577	64.76%
Female	314	35.24%

Key Insight: Male passengers outnumbered female passengers nearly 2:1. This gender imbalance has direct implications for survival pattern analysis.

4.3 Passenger Class Distribution

Class	Count	Percentage	Socioeconomic Level
1st	216	24.24%	Upper class
2nd	184	20.65%	Middle class
3rd	491	55.11%	Lower class

Key Insight: Third-class passengers formed the largest group (>55%), reflecting the socioeconomic composition of early 20th-century transatlantic travel.

4.4 Age & Fare Distributions

- **Age:** Right-skewed distribution. Most passengers were between 20–40 years old. Children and elderly passengers represented a smaller proportion.
- **Fare:** Highly right-skewed. Most passengers paid low fares (median ~£14.45), while a small number paid exceptionally high fares (max £512.33), reflecting socioeconomic disparities.
- **Outliers:** Age shows a few potential outliers representing elderly passengers. Fare exhibits numerous extreme values — both reflect genuine characteristics, not errors.

5. Bivariate Analysis

Bivariate analysis examines relationships between two variables to uncover patterns and associations that may influence passenger survival.

5.1 Survival by Gender

Gender	Did Not Survive	Survived	Survival Rate
Male	468	109	18.9%
Female	81	233	74.2%

Key Insight: Female passengers had a survival rate of ~74% compared to only ~19% for males. This strongly supports the historical 'women and children first' evacuation protocol. Gender is the strongest single predictor of survival in this dataset.

5.2 Survival by Passenger Class

Class	Did Not Survive	Survived	Survival Rate
1st	80	136	63.0%
2nd	97	87	47.3%
3rd	372	119	24.2%

Key Insight: Survival likelihood declined sharply as passenger class decreased: 63% for 1st class, 47% for 2nd, and only 24% for 3rd. This indicates that socioeconomic status significantly influenced access to lifeboats and safety resources.

5.3 Age Distribution by Survival Status

- Children appeared to have relatively better survival outcomes.
- Adult passengers accounted for the majority of fatalities.
- The relationship between age and survival is less pronounced than gender or class.
- Age alone is not a strong determinant of survival.

5.4 Survival by Embarkation Port

Port	Code	Did Not Survive	Survived
Southampton	S	427	219
Cherbourg	C	75	93
Queenstown	Q	47	30

- Southampton had the largest boarding group but also the most fatalities.
- Cherbourg passengers showed a higher relative survival rate, likely due to a higher proportion of 1st-class passengers boarding there.
- Differences in port survival rates may be explained by passenger class composition rather than embarkation location itself.

5.5 Fare by Passenger Class and Survival

- First-class passengers paid substantially higher fares than 2nd and 3rd class.
- Third-class passengers predominantly occupied the low-fare range and experienced the greatest number of fatalities.
- Survivors appear more frequently among high-fare (1st class) passengers, reinforcing the socioeconomic influence on survival.

6. Correlation Analysis

A correlation heatmap was computed across all numerical variables to identify linear associations. Key findings from the correlation matrix:

Variable Pair	Correlation	Interpretation
Survived ↔ Pclass	-0.34	Higher class = better survival chances
Survived ↔ Fare	+0.26	Higher fare = better survival chances
Survived ↔ Age	-0.06	Very weak; age not a major determinant

Pclass ↔ Fare	-0.55	Strong: 1st class paid much higher fares
SibSp ↔ Parch	+0.41	Moderate: larger families travel together
Age ↔ Pclass	-0.34	Older passengers slightly more likely in 1st

Key Insight: Passenger class shows the strongest correlation with survival (-0.34), followed by fare (+0.26). Age has only a negligible relationship with survival. The strong Pclass-Fare correlation (-0.55) confirms these variables together capture socioeconomic status.

7. Summary of Key Findings

Gender	Female passengers had a ~74% survival rate vs ~19% for males, strongly supporting the 'women and children first' evacuation protocol.
Passenger Class	Survival rates declined from 63% (1st) to 47% (2nd) to 24% (3rd). Socioeconomic status was a major factor in survival.
Fare	Higher fares were positively correlated with survival (+0.26), reflecting the advantage of 1st-class accommodations.
Age	Most passengers were young adults (20–40). Age alone was not a strong predictor of survival (-0.06 correlation).
Class Imbalance	62% of passengers did not survive; this imbalance should be addressed in any predictive modelling task.
Missing Data	Cabin was ~77% missing and retained as-is. Age (20% missing) was imputed using the median. Embarked had negligible missingness.
Outliers	Extreme values in Age and Fare represent genuine passenger characteristics and were retained for analysis integrity.
Socioeconomic Impact	Correlation analysis reinforces that passenger class and fare together are the most influential factors in survival outcomes.

8. Conclusion

This project conducted a comprehensive Exploratory Data Analysis of the Titanic dataset using Python libraries including Pandas, Matplotlib, and Seaborn. Through descriptive statistics, data cleaning, and visual analysis, meaningful patterns and relationships were identified.

The analysis demonstrated that **gender, passenger class, and fare** played significant roles in determining survival outcomes, while age exhibited a comparatively weaker influence. The findings highlight the profound impact of socioeconomic status on survival during the Titanic disaster.

This EDA lays a strong foundation for future predictive modelling tasks, such as building classification models to predict passenger survival. The insights derived illustrate how exploratory data analysis can reveal valuable patterns from historical datasets.